

Chiari-like Malformation (COMS) and Syringomyelia

What is COMS?

Caudal occipital malformation syndrome (COMS, or Chiari-like malformation) occurs when the bone at the back of the skull is misshapen and too small. This crowds the back of the brain (cerebellum and brainstem) and blocks normal spinal fluid flow, which can cause fluid buildup in the brain (hydrocephalus) and in the spinal cord (**syrinx**, or syringomyelia).

Many dogs with the malformation are normal. About 90% of Cavalier King Charles Spaniels have it, most without any neurological problems.

Signs

- **"Phantom scratching"** at the ear, neck or shoulder, often without touching the skin
- Neck pain, sensitivity to touch, yelping
- Unusual walking, weakness, wobbliness, dizziness
- Rarely, paralysis

A **syrinx** is a fluid-filled cavity in the spinal cord. The pressure can cause pain, weakness and walking problems. It can occur with COMS or on its own.

Diagnosis

MRI of the brain and neck shows the malformation and any syrinx, and rules out other causes.

Treatment

Not every dog needs treatment. Many live their whole lives with the malformation without problems. Treatment aims to **relieve signs** — with medication or surgery.

Medication

There are few long-term studies. Options fall into three groups:

- **Pain relief:** anti-inflammatories for mild cases; nerve-pain drugs (such as gabapentin) for more severe pain
- **Reducing spinal fluid production:** omeprazole (evidence is limited; not recommended for longer than 8 weeks), or diuretics such as furosemide or acetazolamide (need regular blood tests to check electrolytes and kidneys)
- **Steroids:** may reduce fluid production and pain. About two-thirds of dogs improve, though signs rarely disappear. Long-term use can cause side effects (infection risk, weight gain, skin problems) and blood monitoring is needed.

Many dogs can be managed with medication for some time but may eventually worsen.

Decompressive surgery

Recommended for **ongoing pain or worsening neurological signs**. Part of the skull and its membrane are removed to relieve pressure. In one study, **81% improved**, though 25% needed a second surgery because of scar tissue. Dogs operated on sooner (weeks rather than many months of signs) did better.

Call us if

Scratching or pain increases, weakness develops, or medication side effects appear.